

# Size and Growth of Business Firms: Theoretical Framework and Empirical Evidence

F. Pammolli <sup>a b</sup>, H.E. Stanley <sup>b</sup>, S. Buldyrev <sup>c</sup>, M. Riccaboni <sup>d</sup>, L. Ponta <sup>b e</sup>

<sup>a</sup> Institutions Markets Technologies Institute for Advanced Studies  
Via S. Micheletto 3, 55100 Lucca, Italy  
[f.pammolli@imtlucca.it](mailto:f.pammolli@imtlucca.it)

<sup>b</sup> Center for Polymer Studies and Department of Physics, Boston University,  
Boston, MA 02215  
[hes@bu.edu](mailto:hes@bu.edu)

<sup>c</sup> Department of Physics, Yeshiva University  
500 West 185th Street, New York, NY 10033  
[buldyrev@yu.edu](mailto:buldyrev@yu.edu)

<sup>d</sup> Department of Computer and Management Sciences, University of Trento  
Via Inama, 5, 38100 Trento, Italy  
[massimo.riccaboni@unitn.it](mailto:massimo.riccaboni@unitn.it)

<sup>e</sup> Department of Physics, Politecnico di Torino  
Corso Duca degli Abruzzi 24, 10129 Torino, Italy

We introduce a model of proportional growth to explain the distribution  $P(g)$  of business-firm growth rates. The model predicts that  $P(g)$  is exponential in the central part and depicts an asymptotic power-law behavior in the tails with an exponent  $\zeta = 3$ . Because of data limitations, previous studies in this field have been focusing exclusively on the Laplace shape of the body of the distribution. First, we test the model at different levels of aggregation in the economy, from products to firms to countries, and we find that the predictions of the model agree with empirical size and growth distributions, as well as with size-variance relationships. In particular, the relationship between the size and the variance of firm growth rates is known to follow an approximate power-law behaviour  $\sigma(S) \approx S^{\beta(S)}$  where  $S$  is the firm size and  $\beta(S) \approx 0.2$  is an exponent that weakly depends on  $S$ . Our model can explain this size variance dependence. In general, the model predicts that  $\beta(S)$  must exhibit a crossover from  $\beta(0) = 0$  to  $\beta(\infty) = 1/2$ . For a realistic set of parameters,  $\beta(S)$  is approximately constant and can vary from 0.14 to 0.2 depending on the average number of units in the firm. We test the model with a unique industry-specific database in which firm sales are given in terms of the sum of the sales of all their products. We find that the model is consistent with the empirically observed size-variance relationship.

## Keywords

Sales data, preferential attachment, pharmaceutical industry, distributions